

INETUM

SmartTender AI

Intelligent Automation of the
Tendering Process

Complete Technical Report

Full Guide • Architecture •
Source Code • Deliverables

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 **Prize:** Gadgets, Internships & Opportunity

 **Date:** February 27, 2026



MV



Full Source Code

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Chapter 1

Executive Summary

🎯 Project Vision

SmartTender AI is an intelligent platform that automates the complete tendering lifecycle—from opportunity detection through expert matching to document generation—using Natural Language Processing (NLP), Machine Learning, and Generative AI technologies.

1.1 The Challenge

Organizations responding to public and private tenders face a process that is:

- 🕒 **Time-consuming:** Manual scanning of dozens of platforms daily
- ⚠️ **Error-prone:** Human fatigue leads to missed opportunities
- 🔄 **Repetitive:** Same document formats filled out for every bid
- 👥 **Resource-intensive:** Experts spend 40+ hours/week on administrative tasks

1.2 Our Solution

SmartTender AI addresses each pain point with a dedicated AI module:

Table 1.1: SmartTender AI Modules Overview

Module	Name	Function
🔍 M1	Tender Detection	Automatically discover, filter, and rank relevant tenders
👥 M2	CV Matching	AI-powered matching of expert profiles to requirements
📄 M3	Document Generation	Auto-generate CVs, cover letters, compliance matrices

1.3 Expected Deliverables

✓ Hackathon Deliverables Checklist

1. **Functional MVP** — All three modules implemented with working code
2. **UI/Mockup** — Streamlit dashboard demonstrating full workflow
3. **AI Components:**
 - Keyword extraction using spaCy NER + TF-IDF
 - CV scoring engine with multi-criteria algorithm
 - Template-based document generator with optional LLM enhancement
4. **Demo Video** — 3-minute script included (Section 10)
5. **Technical Summary** — This complete report with architecture, data flow, AI techniques

Chapter 2

Problem Analysis

2.1 Current Workflow Pain Points

Table 2.1: Time Distribution in Traditional Tendering Process

Phase	Description	Time (%)	Hours/Week
Detection	Scanning TUNEPS, TED, SAM.gov, portals daily	30%	12h
Analysis	Reading 50–200 page documents, extracting requirements	25%	10h
Matching	Selecting experts whose CVs fit technical criteria	25%	10h
Generation	Producing administrative & technical response docs	20%	8h
Total Weekly Effort		100%	40h

2.2 Stakeholder Needs Analysis

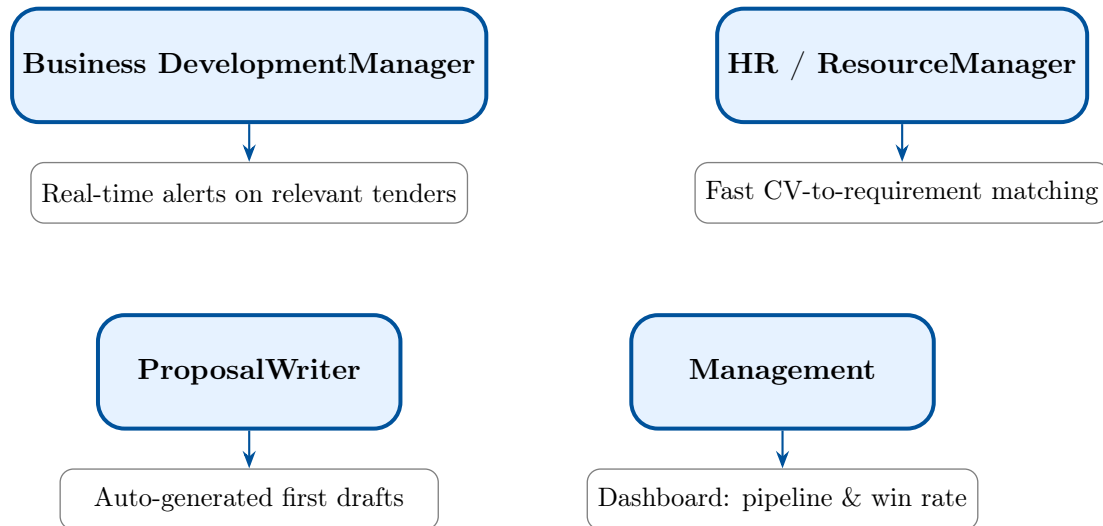


Figure 2.1: Stakeholder Needs Mapping

2.3 Key Performance Indicators (KPIs)

Table 2.2: Target KPIs for SmartTender AI

inetumblue!20 Metric	Current	Target	Improvement
Time to identify relevant tender	2 hours	5 minutes	96% ↓
CV matching accuracy	60%	90%	50% ↑
Document preparation time	4 hours	30 minutes	87% ↓
Missed tender opportunities	20%	2%	90% ↓

Chapter 3

Solution Architecture

3.1 High-Level System Architecture

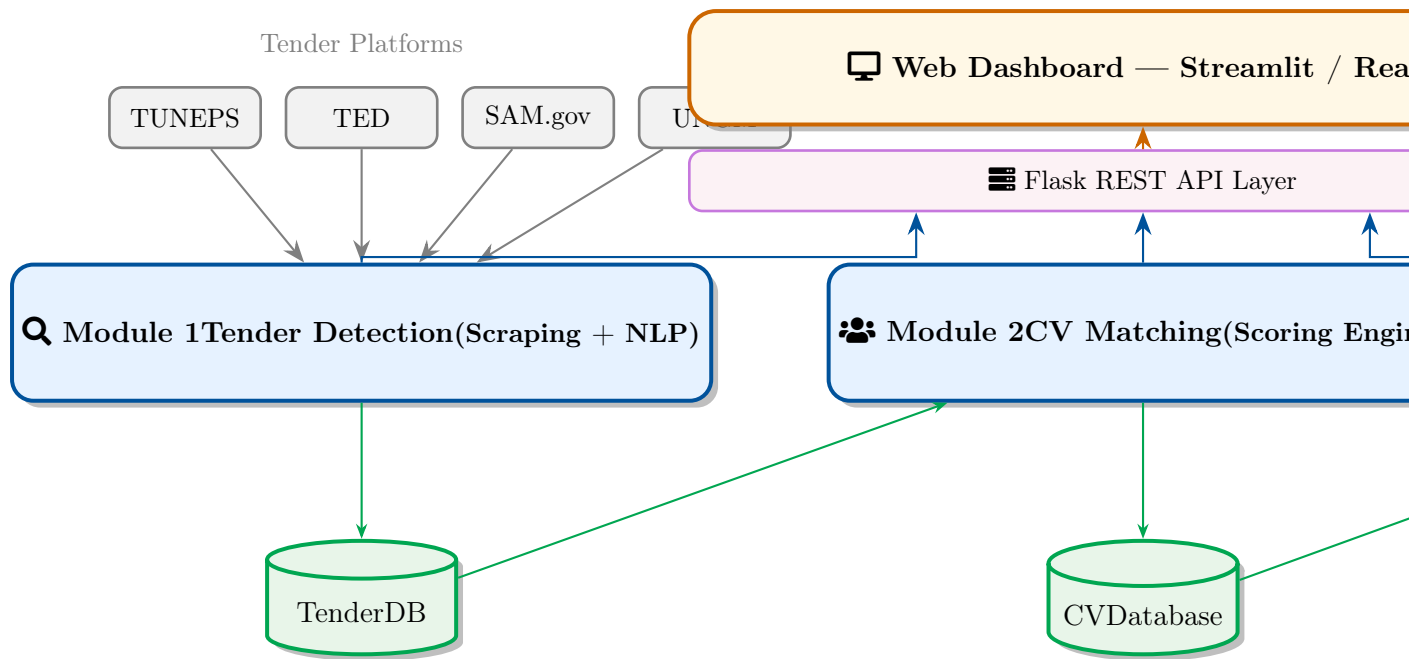


Figure 3.1: SmartTender AI — High-Level System Architecture

3.2 Data Flow Diagram

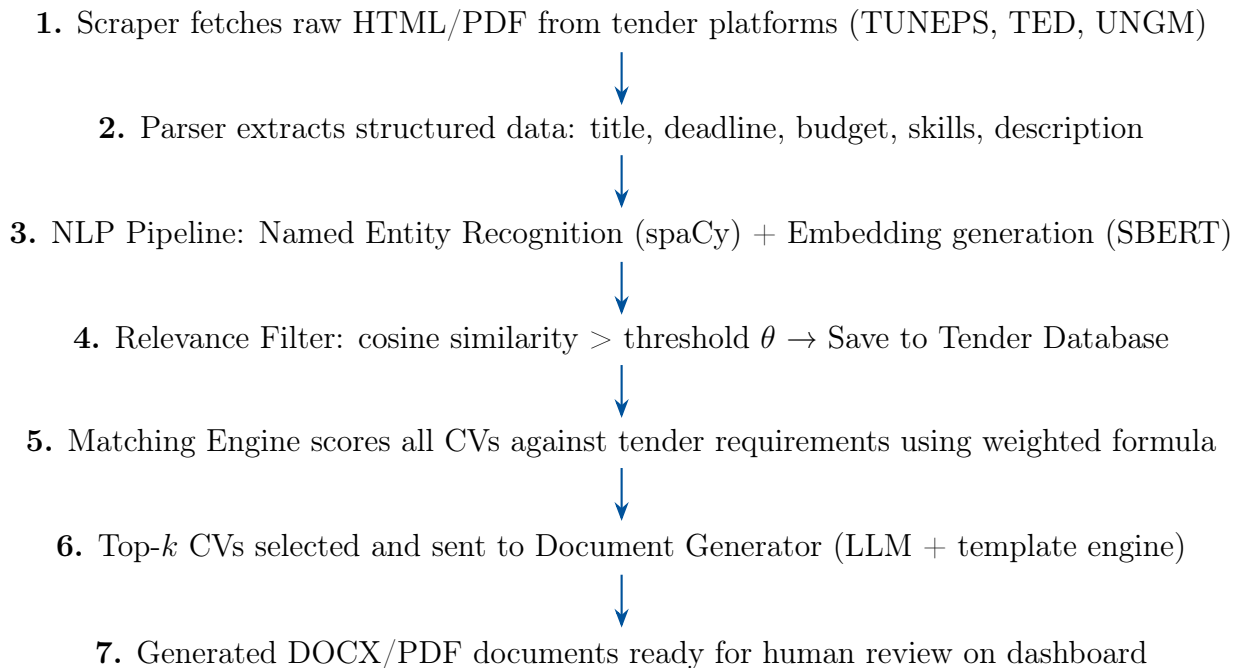


Figure 3.2: End-to-End Data Flow

3.3 Technology Stack

Table 3.1: Technology Stack Selection

Layer	Technology	Justification
Backend API	Python + Flask	Rapid prototyping, rich AI/ML ecosystem
NLP	spaCy, HuggingFace, Sentence-BERT	State-of-art NER, embeddings
LLM	OpenAI GPT-4 / Ollama + Mistral	Generative text capability
Scraping	BeautifulSoup, Scrapy, Selenium	Multi-platform crawling
Frontend	Streamlit (MVP) / React (prod)	Fast UI for demonstrations
Database	SQLite (MVP) $\rightarrow$ PostgreSQL	Simple to scalable
Vector Store	FAISS / ChromaDB	Fast similarity search
Doc Generation	python-docx, Jinja2, WeasyPrint	Template-based DOCX/PDF export

Chapter 4

Module 1: Smart Tender Detection

4.1 Objective

🎯 Module 1 Goal

Automatically discover, filter, and rank tenders relevant to the company's domain and capabilities using NLP-based keyword extraction and semantic similarity scoring.

4.2 Technical Approach

1. **Web Scraping:** Scrapy/BeautifulSoup/Selenium crawl TUNEPS, TED, SAM.gov, UNGM
2. **Keyword Extraction:** spaCy NER + TF-IDF extract domain, skills, budget, deadline
3. **Relevance Scoring:** Cosine similarity between company profile embedding and tender embedding
4. **Alert System:** Email/Slack notification when score exceeds threshold

4.3 Mathematical Formulation

4.3.1 Relevance Score

The relevance score between a tender t and company profile c is computed as:

$$\text{score}(t, c) = \frac{\vec{e}_t \cdot \vec{e}_c}{\|\vec{e}_t\| \cdot \|\vec{e}_c\|} \quad (4.1)$$

where:

- $\vec{e}_t$ is the Sentence-BERT embedding of the tender description
- $\vec{e}_c$ is the Sentence-BERT embedding of the company profile

4.3.2 TF-IDF for Keyword Extraction

$$\text{TF-IDF}(t, d, D) = \text{tf}(t, d) \times \log \frac{|D|}{|\{d \in D : t \in d\}|} \quad (4.2)$$

4.4 Complete Source Code

4.4.1 File: module1_tender_detection.py

```

1  """
2  SmartTender AI - Module 1: Smart Tender Detection
3  =====
4  Features:
5  - Keyword extraction (spaCy + TF-IDF)
6  - Semantic embedding (Sentence-BERT)
7  - Relevance scoring (cosine similarity)
8  - Platform simulation (mock scraper)
9
10 Author: SmartTender AI Team
11 Date: 2025
12 """
13
14 import re
15 import spacy
16 import numpy as np
17 from sklearn.feature_extraction.text import TfidfVectorizer
18 from sentence_transformers import SentenceTransformer
19 from sklearn.metrics.pairwise import cosine_similarity
20 from typing import Dict, List, Any, Optional
21
22 # =====
23 # LOAD NLP MODELS
24 # =====
25 print("[Module 1] Loading NLP models...")
26
27 # Load spaCy model for NER
28 try:
29     nlp = spacy.load("en_core_web_sm")
30 except OSError:
31     print(" -> Downloading spaCy model...")
32     import subprocess
33     subprocess.run(["python", "-m", "spacy", "download", "en_core_web_sm"])
34     nlp = spacy.load("en_core_web_sm")
35
36 # Load Sentence-BERT for semantic embeddings
37 sbert_model = SentenceTransformer("all-MiniLM-L6-v2")
38 print("[Module 1] Models loaded successfully!")
39
40
41 # =====
42 # DEFAULT COMPANY PROFILE
43 # =====
44 DEFAULT_COMPANY_PROFILE = {
45     "name": "Inetum Tunisie",
46     "description": ""
47     Inetum is a leading IT services company specializing in

```

```

48         digital transformation, cloud computing, ERP implementation,
49         AI/ML solutions, custom software development, cybersecurity,
50         and IT consulting. We operate across Europe, Africa, and the
51         Middle East with expertise in Python, Java, AWS, Azure, SAP,
52         Odoo, data analytics, and agile project management.
53     """
54     "domains": [
55         "IT Services", "Cloud Computing", "ERP",
56         "AI / Machine Learning", "Cybersecurity",
57         "Digital Transformation", "Software Development"
58     ],
59     "key_skills": [
60         "Python", "Java", "AWS", "Azure", "Docker", "Kubernetes",
61         "SAP", "Odoo", "NLP", "Machine Learning", "Agile/Scrum",
62         "REST API", "Microservices", "PostgreSQL", "React"
63     ]
64 }
65
66
67 # =====
68 # TENDER DETECTOR CLASS
69 # =====
70 class TenderDetector:
71     """
72     Smart Tender Detection Engine
73
74     This class provides functionality to:
75     1. Extract keywords from tender documents
76     2. Compute relevance scores between tenders and company profile
77     3. Rank and filter tenders by relevance
78     """
79
80     def __init__(
81         self,
82         company_profile: Optional[Dict] = None,
83         threshold: float = 0.3
84     ):
85         """
86         Initialize the TenderDetector.
87
88         Args:
89             company_profile: Dictionary containing company
90                             information
91             threshold: Minimum relevance score (0-1) to consider
92                       relevant
93         """
94         self.company = company_profile or DEFAULT_COMPANY_PROFILE
95         self.threshold = threshold
96
97         # Pre-compute company profile embedding for efficiency
98         self.company_embedding = sbert_model.encode(
99             [self.company["description"]]
100         )

```

```

99
100 # =====
101 # KEYWORD EXTRACTION
102 # =====
103 def extract_keywords(
104     self,
105     text: str,
106     top_n: int = 15
107 ) -> Dict[str, List[str]]:
108     """
109     Extract keywords from tender text using multiple NLP
110     techniques.
111
112     Args:
113         text: The tender description text
114         top_n: Number of top keywords to extract
115
116     Returns:
117         Dictionary containing:
118         - entities: Named entities (organizations, dates, etc.)
119         - noun_chunks: Noun phrases
120         - tfidf_keywords: TF-IDF extracted terms
121         - tech_skills: Technical skills (pattern-matched)
122     """
123     doc = nlp(text)
124
125     # 1. Named Entity Recognition
126     entity_labels = ("ORG", "GPE", "MONEY", "DATE",
127                     "PRODUCT", "NORP", "FAC")
128     entities = list(set([
129         ent.text.strip()
130         for ent in doc.ents
131         if ent.label_ in entity_labels and len(ent.text.strip())
132         > 2
133     ]))
134
135     # 2. Noun Chunks (noun phrases)
136     chunks = list(set([
137         chunk.text.strip().lower()
138         for chunk in doc.noun_chunks
139         if len(chunk.text.strip()) > 2 and len(chunk.text.split())
140         <= 4
141     ]))[:20]
142
143     # 3. TF-IDF Keywords
144     sentences = [
145         sent.text for sent in doc.sents
146         if len(sent.text.strip()) > 10
147     ]
148     tfidf_keywords = []
149     if sentences:
150         try:
151             tfidf = TfidfVectorizer(

```

```

149         stop_words="english",
150         max_features=top_n,
151         ngram_range=(1, 2),
152         min_df=1
153     )
154     tfidf.fit_transform(sentences)
155     tfidf_keywords = tfidf.get_feature_names_out().tolist()
156     except ValueError:
157         pass # Handle empty vocabulary
158
159 # 4. Technical Skills (Pattern-Based)
160 tech_patterns = [
161     r'\b(Python|Java|JavaScript|TypeScript|C\+\+|C#|Go|Rust|
162         Ruby|PHP)\b',
163     r'\b(React|Angular|Vue\.js|Node\.js|Django|Flask|Spring|
164         Laravel)\b',
165     r'\b(AWS|Azure|GCP|Google Cloud|Docker|Kubernetes|K8s|
166         Terraform)\b',
167     r'\b(PostgreSQL|MySQL|MongoDB|Redis|Elasticsearch|Oracle|
168         SQL Server)\b',
169     r'\b(SAP|Odoo|Salesforce|ServiceNow|Dynamics)\b',
170     r'\b(NLP|ML|AI|TensorFlow|PyTorch|Scikit-learn|Deep
171         Learning)\b',
172     r'\b(REST\*API|GraphQL|Microservices|CI/CD|DevOps|MLOps)
173         \b',
174     r'\b(Agile|Scrum|Kanban|PMP|ITIL|PRINCE2|Jira)\b',
175     r'\b(ISO\s*27001|GDPR|OWASP|CISSP|CEH|SOC2|HIPAA)\b',
176     r'\b(Linux|Windows Server|Unix|Shell|Bash|PowerShell)\b',
177 ]
178 tech_skills = []
179 for pattern in tech_patterns:
180     matches = re.findall(pattern, text, re.IGNORECASE)
181     tech_skills.extend([m.strip() for m in matches])
182 tech_skills = list(set(tech_skills))
183
184 return {
185     "entities": entities,
186     "noun_chunks": chunks[:15],
187     "tfidf_keywords": tfidf_keywords,
188     "tech_skills": tech_skills
189 }
190
191 # =====
192 # RELEVANCE SCORING
193 # =====
194 def compute_relevance(self, tender: Dict) -> float:
195     """
196     Compute relevance score between tender and company profile.
197
198     The score is a weighted combination of:
199     - Semantic similarity (45%): SBERT embeddings cosine
200       similarity

```



```
194     - Skill overlap (35%): Jaccard-like skill matching
195     - Domain match (20%): Category alignment
196
197     Args:
198         tender: Dictionary containing tender information
199
200     Returns:
201         Relevance score as percentage (0-100)
202     """
203     # 1. Semantic Similarity (Sentence-BERT)
204     tender_text = f"{tender.get('title', '')}. {tender.get('description', '')}"
205     tender_embedding = sbert_model.encode([tender_text])
206     semantic_sim = cosine_similarity(
207         tender_embedding,
208         self.company_embedding
209     )[0][0]
210
211     # 2. Skill Overlap Score
212     tender_skills = set([
213         s.lower().strip()
214         for s in tender.get("required_skills", [])
215     ])
216     company_skills = set([
217         s.lower().strip()
218         for s in self.company["key_skills"]
219     ])
220
221     if tender_skills:
222         skill_overlap = len(tender_skills & company_skills) / len(
223             tender_skills)
224     else:
225         skill_overlap = 0.5 # Neutral if no skills specified
226
227     # 3. Domain Match Score
228     tender_category = tender.get("category", "").lower()
229     domain_match = any(
230         domain.lower() in tender_category or
231         tender_category in domain.lower()
232         for domain in self.company["domains"]
233     )
234     domain_score = 1.0 if domain_match else 0.3
235
236     # 4. Weighted Final Score
237     weights = {
238         "semantic": 0.45,
239         "skills": 0.35,
240         "domain": 0.20
241     }
242
243     final_score = (
244         weights["semantic"] * semantic_sim +
245         weights["skills"] * skill_overlap +
```

```
245         weights["domain"] * domain_score
246     )
247
248     return round(float(final_score * 100), 2)
249
250     # =====
251     # ANALYZE TENDERS
252     # =====
253     def analyze_tenders(
254         self,
255         tenders: List[Dict]
256     ) -> List[Dict[str, Any]]:
257         """
258         Analyze a list of tenders and return ranked results.
259
260         Args:
261             tenders: List of tender dictionaries
262
263         Returns:
264             List of analysis results, sorted by relevance score
265         """
266         results = []
267
268         for tender in tenders:
269             # Compute relevance score
270             score = self.compute_relevance(tender)
271
272             # Extract keywords
273             keywords = self.extract_keywords(
274                 tender.get("description", "")
275             )
276
277             # Build result object
278             result = {
279                 "id": tender.get("id", ""),
280                 "title": tender.get("title", ""),
281                 "platform": tender.get("platform", "Unknown"),
282                 "deadline": tender.get("deadline", "N/A"),
283                 "budget": tender.get("budget", "N/A"),
284                 "category": tender.get("category", "Unknown"),
285                 "relevance_score": score,
286                 "is_relevant": score >= (self.threshold * 100),
287                 "extracted_keywords": keywords,
288                 "required_skills": tender.get("required_skills", []),
289                 "analysis_timestamp": str(np.datetime64('now'))
290             }
291             results.append(result)
292
293             # Sort by relevance score (descending)
294             results.sort(key=lambda x: x["relevance_score"], reverse=True)
295
296         return results
```

```

297
298 # =====
299 # MOCK SCRAPER (for demonstration)
300 # =====
301 @staticmethod
302 def mock_scrape(platform: str = "TUNEPS") -> List[Dict]:
303     """
304     Simulate scraping a tender platform.
305     In production, this would use actual web scraping.
306
307     Args:
308         platform: Name of the platform to scrape
309
310     Returns:
311         List of tender dictionaries
312     """
313     # Import sample data
314     from utils import load_json
315     all_tenders = load_json("sample_tenders.json")
316
317     # Filter by platform if specified
318     if platform.upper() != "ALL":
319         filtered = [
320             t for t in all_tenders
321             if t.get("platform", "").upper() == platform.upper()
322         ]
323         return filtered if filtered else all_tenders
324
325     return all_tenders
326
327
328 # =====
329 # STANDALONE EXECUTION
330 # =====
331 if __name__ == "__main__":
332     import json
333     from pathlib import Path
334
335     # Load sample data
336     data_dir = Path(__file__).parent.parent / "data"
337     sample_file = data_dir / "sample_tenders.json"
338
339     if sample_file.exists():
340         with open(sample_file, "r", encoding="utf-8") as f:
341             tenders = json.load(f)
342     else:
343         # Fallback sample tender
344         tenders = [{
345             "id": "T001",
346             "title": "Cloud ERP System Development",
347             "description": "Development of cloud-based ERP using

```

```

348         "required_skills": ["Python", "AWS", "Docker", "
349             PostgreSQL"],
350         "platform": "TUNEPS",
351         "deadline": "2025-03-01",
352         "category": "IT Services"
353     }]
354
355     # Initialize detector
356     detector = TenderDetector(threshold=0.3)
357
358     # Analyze tenders
359     results = detector.analyze_tenders(tenders)
360
361     # Display results
362     print("\n" + "=" * 70)
363     print(" SMARTTENDER AI - TENDER DETECTION RESULTS")
364     print("=" * 70)
365
366     for r in results:
367         status = "RELEVANT" if r["is_relevant"] else "LOW MATCH"
368         icon = "[OK]" if r["is_relevant"] else "--"
369
370         print(f"\n{icon} {r['title']}")
371         print(f"    Score: {r['relevance_score']}% | Status: {status}"
372             ")")
373         print(f"    Platform: {r['platform']} | Deadline: {r['"
374             deadline']}'")
375         print(f"    Tech Skills Found: {', '.join(r['"
376             extracted_keywords']}'tech_skills'][:5])}")
377
378     print("\n" + "=" * 70)
379     relevant_count = sum(1 for r in results if r["is_relevant"])
380     print(f" Summary: {relevant_count}/{len(results)} tenders are"
381         relevant")
382     print("=" * 70)

```

Listing 4.1: Module 1: Smart Tender Detection — Complete Implementation

Chapter 5

Module 2: AI CV Matching

5.1 Objective

🎯 Module 2 Goal

Given a tender's technical requirements, rank internal CVs by fitness score, identify matched and missing skills, and provide actionable recommendations for team composition.

5.2 Scoring Algorithm

The CV matching score is computed using a weighted multi-criteria formula:

$$S_{cv} = w_1 \cdot \text{SkillMatch} + w_2 \cdot \text{ExpScore} + w_3 \cdot \text{CertBonus} + w_4 \cdot \text{DomainSim} \quad (5.1)$$

where the default weights are:

- $w_1 = 0.35$ — Skill Match (Jaccard similarity)
- $w_2 = 0.20$ — Experience Score (years normalized)
- $w_3 = 0.15$ — Certification Bonus
- $w_4 = 0.30$ — Domain Similarity (semantic)

5.2.1 Component Formulas

Skill Match (Jaccard Index):

$$\text{SkillMatch} = \frac{|R \cap C|}{|R|} \quad (5.2)$$

where R = required skills, C = candidate skills.

Experience Score:

$$\text{ExpScore} = \min \left(\frac{\text{CV Years}}{\text{Required Years}}, 1.5 \right) / 1.5 \quad (5.3)$$

Domain Similarity:

$$\text{DomainSim} = \text{cosine}(\vec{e}_{\text{tender}}, \vec{e}_{\text{cv}}) \quad (5.4)$$

5.3 Complete Source Code

5.3.1 File: module2_cv_matching.py

```

1  """
2  SmartTender AI - Module 2: AI CV Matching
3  =====
4  Features:
5  - Multi-criteria scoring (skills, experience, certs, domain)
6  - Semantic similarity using Sentence-BERT
7  - Gap analysis for team composition
8  - Ranking engine with recommendations
9
10 Author: SmartTender AI Team
11 Date: 2025
12 """
13
14 import numpy as np
15 from sentence_transformers import SentenceTransformer
16 from sklearn.metrics.pairwise import cosine_similarity
17 from typing import Dict, List, Any, Optional, Tuple
18
19 # =====
20 # LOAD EMBEDDING MODEL
21 # =====
22 print("[Module 2] Loading embedding model...")
23 model = SentenceTransformer("all-MiniLM-L6-v2")
24 print("[Module 2] Model loaded successfully!")
25
26
27 # =====
28 # CV MATCHER CLASS
29 # =====
30 class CVMatcher:
31     """
32     AI-Powered CV Matching Engine
33
34     Matches candidate CVs against tender requirements using:
35     1. Skill overlap (Jaccard-based)
36     2. Experience scoring (normalized years)
37     3. Certification matching
38     4. Semantic domain similarity (SBERT)
39     """
40
41     def __init__(self, weights: Optional[Dict[str, float]] = None):
42         """
43         Initialize the CV Matcher.
44
45         Args:
46             weights: Custom weights for scoring components
47         """
48         self.weights = weights or {

```

```

49         "skill_match": 0.35,
50         "experience": 0.20,
51         "certification": 0.15,
52         "domain_similarity": 0.30
53     }
54
55     # =====
56     # INDIVIDUAL SCORE COMPONENTS
57     # =====
58     @staticmethod
59     def skill_match_score(
60         required_skills: List[str],
61         cv_skills: List[str]
62     ) -> Tuple[float, List[str], List[str]]:
63         """
64         Calculate skill match using set operations.
65
66         Args:
67             required_skills: Skills required by the tender
68             cv_skills: Skills possessed by the candidate
69
70         Returns:
71             Tuple of (score, matched_skills, missing_skills)
72         """
73         # Normalize to lowercase for comparison
74         req = set(s.lower().strip() for s in required_skills if s)
75         cv = set(s.lower().strip() for s in cv_skills if s)
76
77         if not req:
78             return 0.0, [], list(cv)
79
80         matched = req & cv
81         missing = req - cv
82
83         # Score based on coverage of required skills
84         score = len(matched) / len(req)
85
86         return round(score, 4), list(matched), list(missing)
87
88     @staticmethod
89     def experience_score(
90         required_years: int,
91         cv_years: int
92     ) -> float:
93         """
94         Calculate experience score with diminishing returns.
95
96         Args:
97             required_years: Minimum years required
98             cv_years: Years of experience on CV
99
100         Returns:
101             Normalized score (0 to 1)

```

```

102         """
103         if required_years <= 0:
104             return 1.0
105
106         # Allow up to 50% bonus for exceeding requirements
107         ratio = cv_years / required_years
108         capped_ratio = min(ratio, 1.5)
109
110         return round(capped_ratio / 1.5, 4)
111
112     @staticmethod
113     def certification_score(
114         required_certs: List[str],
115         cv_certs: List[str]
116     ) -> Tuple[float, List[str], List[str]]:
117         """
118         Calculate certification match score.
119
120         Args:
121             required_certs: Certifications required/preferred
122             cv_certs: Certifications held by candidate
123
124         Returns:
125             Tuple of (score, matched_certs, missing_certs)
126         """
127         req = set(c.lower().strip() for c in required_certs if c)
128         cv = set(c.lower().strip() for c in cv_certs if c)
129
130         if not req:
131             # No certifications required - give full score
132             return 1.0, list(cv), []
133
134         matched = req & cv
135         missing = req - cv
136
137         score = len(matched) / len(req)
138
139         return round(score, 4), list(matched), list(missing)
140
141     @staticmethod
142     def domain_similarity_score(
143         tender_description: str,
144         cv_summary: str
145     ) -> float:
146         """
147         Calculate semantic similarity between tender and CV.
148
149         Uses Sentence-BERT embeddings for semantic understanding.
150
151         Args:
152             tender_description: Tender description text
153             cv_summary: CV professional summary text
154

```



```

155     Returns:
156         Cosine similarity score (0 to 1)
157     """
158     if not tender_description or not cv_summary:
159         return 0.5 # Neutral score if text missing
160
161     # Generate embeddings
162     emb_tender = model.encode([tender_description])
163     emb_cv = model.encode([cv_summary])
164
165     # Compute cosine similarity
166     sim = cosine_similarity(emb_tender, emb_cv)[0][0]
167
168     return round(float(sim), 4)
169
170 # =====
171 # FULL CV SCORING
172 # =====
173 def score_cv(
174     self,
175     tender: Dict,
176     cv: Dict
177 ) -> Dict[str, Any]:
178     """
179     Compute full matching score for one CV against a tender.
180
181     Args:
182         tender: Tender requirements dictionary
183         cv: CV data dictionary
184
185     Returns:
186         Comprehensive scoring result dictionary
187     """
188     # 1. Skill Match
189     skill_score, matched_skills, missing_skills = self.
190         skill_match_score(
191             tender.get("required_skills", []),
192             cv.get("skills", [])
193         )
194
195     # 2. Experience Score
196     exp_score = self.experience_score(
197         tender.get("min_experience_years", 0),
198         cv.get("years_experience", 0)
199     )
200
201     # 3. Certification Score
202     cert_score, matched_certs, missing_certs = self.
203         certification_score(
204             tender.get("required_certifications", []),
205             cv.get("certifications", [])

```

```

206     # 4. Domain Similarity
207     domain_score = self.domain_similarity_score(
208         tender.get("description", ""),
209         cv.get("summary", "")
210     )
211
212     # Calculate Weighted Total
213     total = (
214         self.weights["skill_match"] * skill_score +
215         self.weights["experience"] * exp_score +
216         self.weights["certification"] * cert_score +
217         self.weights["domain_similarity"] * domain_score
218     )
219
220     # Build result dictionary
221     return {
222         "cv_id": cv.get("id", ""),
223         "cv_name": cv.get("name", "Unknown"),
224         "cv_title": cv.get("title", ""),
225         "cv_email": cv.get("email", ""),
226         "total_score": round(total * 100, 2),
227         "breakdown": {
228             "skill_match": round(skill_score * 100, 2),
229             "experience": round(exp_score * 100, 2),
230             "certification": round(cert_score * 100, 2),
231             "domain_similarity": round(domain_score * 100, 2)
232         },
233         "matched_skills": matched_skills,
234         "missing_skills": missing_skills,
235         "matched_certifications": matched_certs,
236         "missing_certifications": missing_certs,
237         "availability": cv.get("availability", "Unknown"),
238         "recommendation": self._generate_recommendation(total)
239     }
240
241     def _generate_recommendation(self, score: float) -> str:
242         """Generate human-readable recommendation based on score."""
243         if score >= 0.75:
244             return "Excellent Match - Highly Recommended"
245         elif score >= 0.55:
246             return "Good Match - Consider with minor gaps"
247         elif score >= 0.35:
248             return "Partial Match - Significant gaps exist"
249         else:
250             return "Poor Match - Not recommended"
251
252     # =====
253     # RANK ALL CVS
254     # =====
255     def rank_cvs(
256         self,
257         tender: Dict,
258         cvs: List[Dict],

```

```

259         top_k: Optional[int] = None
260     ) -> List[Dict[str, Any]]:
261         """
262         Rank all CVs against a tender by matching score.
263
264         Args:
265             tender: Tender requirements
266             cvs: List of CV dictionaries
267             top_k: Limit results to top K (optional)
268
269         Returns:
270             Sorted list of scoring results
271         """
272         results = [self.score_cv(tender, cv) for cv in cvs]
273
274         # Sort by total score descending
275         results.sort(key=lambda x: x["total_score"], reverse=True)
276
277         # Limit to top_k if specified
278         if top_k and top_k > 0:
279             results = results[:top_k]
280
281         return results
282
283     # =====
284     # TEAM GAP ANALYSIS
285     # =====
286     def gap_analysis(
287         self,
288         tender: Dict,
289         cvs: List[Dict]
290     ) -> Dict[str, Any]:
291         """
292         Analyze skill coverage across all available CVs.
293
294         Helps identify skills that need external recruitment.
295
296         Args:
297             tender: Tender requirements
298             cvs: List of all available CVs
299
300         Returns:
301             Gap analysis dictionary
302         """
303         # Collect all required skills
304         all_required = set(
305             s.lower().strip()
306             for s in tender.get("required_skills", [])
307         )
308
309         # Collect all skills from all CVs
310         all_available = set()
311         for cv in cvs:

```

```

312         for skill in cv.get("skills", []):
313             all_available.add(skill.lower().strip())
314
315         # Calculate coverage
316         covered = all_required & all_available
317         uncovered = all_required - all_available
318         coverage_pct = len(covered) / max(len(all_required), 1)
319
320         return {
321             "total_required_skills": len(all_required),
322             "total_available_skills": len(all_available),
323             "covered_skills": list(covered),
324             "uncovered_skills": list(uncovered),
325             "team_coverage_percentage": round(coverage_pct * 100, 2),
326             "staffing_recommendation": (
327                 "Team is well-equipped" if coverage_pct >= 0.8
328                 else "Consider external recruitment for: " + ", ".join(uncovered)
329             )
330         }
331
332
333 # =====
334 # STANDALONE EXECUTION
335 # =====
336 if __name__ == "__main__":
337     import json
338     from pathlib import Path
339
340     # Load sample data
341     data_dir = Path(__file__).parent.parent / "data"
342
343     # Load tenders
344     with open(data_dir / "sample_tenders.json", "r") as f:
345         tenders = json.load(f)
346
347     # Load CVs
348     with open(data_dir / "sample_cvs.json", "r") as f:
349         cvs = json.load(f)
350
351     # Initialize matcher
352     matcher = CVMatcher()
353
354     # Process each tender
355     for tender in tenders:
356         print("\n" + "=" * 70)
357         print(f" TENDER: {tender['title']}")
358         print("=" * 70)
359
360         # Rank CVs
361         rankings = matcher.rank_cvs(tender, cvs)
362
363         # Display rankings

```

```

364     for i, r in enumerate(rankings, 1):
365         medal = ["1st", "2nd", "3rd"][i-1] if i <= 3 else f"{i}th"
366         print(f"\n {medal}: {r['cv_name']} ({r['cv_title']})")
367         print(f"         Total Score: {r['total_score']}%")
368         print(f"         Skills: {r['breakdown']['skill_match']}% | "
369               f"Exp: {r['breakdown']['experience']}% | "
370               f"Certs: {r['breakdown']['certification']}% | "
371               f"Domain: {r['breakdown']['domain_similarity']}%")
372         print(f"         -> {r['recommendation']}")
373         if r['missing_skills']:
374             print(f"         Missing: {', '.join(r['missing_skills'][:5])}")
375
376     # Gap analysis
377     gap = matcher.gap_analysis(tender, cvs)
378     print(f"\n Team Coverage: {gap['team_coverage_percentage']}%")
379     if gap["uncovered_skills"]:
380         print(f" Uncovered Skills: {', '.join(gap['uncovered_skills'])}")
381
382     print("\n" + "=" * 70)
383     print(" CV MATCHING COMPLETE")
384     print("=" * 70)

```

Listing 5.1: Module 2: AI CV Matching — Complete Implementation

Chapter 6

Module 3: AI Document Generation

6.1 Objective

🎯 Module 3 Goal

Auto-generate standardized tender response documents including formatted CVs, cover letters, and compliance matrices using template engines and optional LLM enhancement.

6.2 Document Types

Table 6.1: Generated Document Types

Document	Format	Purpose
Expert CV	DOCX/PDF	Professional CV in client-required format
Cover Letter	DOCX/PDF	Formal response letter to tender
Compliance Matrix	DOCX/XLSX	Requirements traceability
Executive Summary	DOCX/PDF	High-level proposal overview

6.3 Complete Source Code

6.3.1 File: module3_doc_generation.py

```
1  """
2  SmartTender AI - Module 3: AI Document Generation
3  =====
4  Features:
5  - Template-based CV generation (DOCX)
6  - Cover letter generation
7  - Compliance matrix generation
8  - Optional LLM-enhanced narrative text
9
10 Author: SmartTender AI Team
11 Date: 2025
12 """
13
14 import os
15 from datetime import datetime
16 from pathlib import Path
17 from typing import Dict, List, Any, Optional
18
```

```

19 # Check for python-docx availability
20 try:
21     from docx import Document
22     from docx.shared import Inches, Pt, Cm, RGBColor
23     from docx.enum.text import WD_ALIGN_PARAGRAPH
24     from docx.enum.table import WD_TABLE_ALIGNMENT
25     from docx.enum.style import WD_STYLE_TYPE
26     HAS_DOCX = True
27 except ImportError:
28     HAS_DOCX = False
29     print("[Warning] python-docx not installed. DOCX generation
        disabled.")
30     print("        Install with: pip install python-docx")
31
32 # Output directory
33 OUTPUT_DIR = Path(__file__).parent.parent / "output"
34 OUTPUT_DIR.mkdir(exist_ok=True)
35
36
37 # =====
38 # DOCUMENT GENERATOR CLASS
39 # =====
40 class DocumentGenerator:
41     """
42     AI-Powered Document Generator for Tender Responses
43
44     Generates:
45     1. Professional CVs in DOCX format
46     2. Cover letters tailored to tender
47     3. Compliance matrices
48     4. Optional LLM-enhanced content
49     """
50
51     def __init__(
52         self,
53         use_llm: bool = False,
54         llm_api_key: Optional[str] = None
55     ):
56         """
57         Initialize the Document Generator.
58
59         Args:
60             use_llm: Whether to use LLM for text enhancement
61             llm_api_key: API key for OpenAI (if using LLM)
62         """
63         self.use_llm = use_llm
64         self.llm_api_key = llm_api_key
65         self.output_dir = OUTPUT_DIR
66
67     # =====
68     # CV DOCUMENT GENERATION
69     # =====
70     def generate_cv_docx(

```

```

71         self,
72         cv_data: Dict,
73         tender_data: Optional[Dict] = None
74     ) -> str:
75         """
76         Generate a professional CV in DOCX format.
77
78         Args:
79             cv_data: CV information dictionary
80             tender_data: Optional tender for contextual enhancement
81
82         Returns:
83             Path to generated DOCX file
84         """
85         if not HAS_DOCX:
86             return self._generate_cv_text(cv_data)
87
88         # Create new document
89         doc = Document()
90
91         # ===== STYLES =====
92         style = doc.styles['Normal']
93         font = style.font
94         font.name = 'Calibri'
95         font.size = Pt(11)
96
97         # ===== HEADER - NAME =====
98         heading = doc.add_heading(level=0)
99         heading.alignment = WD_ALIGN_PARAGRAPH.LEFT
100        run = heading.add_run(cv_data.get("name", "Candidate Name"))
101        run.font.size = Pt(26)
102        run.font.color.rgb = RGBColor(0, 82, 155) # Inetum blue
103        run.font.bold = True
104
105        # ===== SUBTITLE - TITLE =====
106        subtitle = doc.add_paragraph()
107        subtitle.alignment = WD_ALIGN_PARAGRAPH.LEFT
108        run = subtitle.add_run(cv_data.get("title", "Professional
109                                Title"))
110        run.font.size = Pt(14)
111        run.font.color.rgb = RGBColor(100, 100, 100)
112        run.font.italic = True
113
114        # ===== CONTACT INFO =====
115        contact = doc.add_paragraph()
116        contact_text = f>Email: {cv_data.get('email', 'email@example.
117                                com')}
118        if cv_data.get("languages"):
119            contact_text += f" | Languages: {'', '.join(cv_data['
120                                languages'])}
121        if cv_data.get("availability"):
122            contact_text += f" | Status: {cv_data['availability']}
123        run = contact.add_run(contact_text)

```



```

121     run.font.size = Pt(10)
122     run.font.color.rgb = RGBColor(80, 80, 80)
123
124     # Horizontal line
125     doc.add_paragraph("_" * 80)
126
127     # ===== PROFESSIONAL SUMMARY =====
128     doc.add_heading("Professional Summary", level=1)
129     summary_text = cv_data.get("summary", "No summary provided.")
130
131     # Optionally enhance with LLM
132     if self.use_llm and tender_data:
133         summary_text = self._enhance_with_llm(
134             summary_text, tender_data, "summary"
135         )
136
137     doc.add_paragraph(summary_text)
138
139     # ===== TECHNICAL SKILLS =====
140     doc.add_heading("Technical Skills", level=1)
141     skills = cv_data.get("skills", [])
142
143     if skills:
144         # Create table for skills (3 columns)
145         num_cols = 3
146         num_rows = (len(skills) + num_cols - 1) // num_cols
147         table = doc.add_table(rows=num_rows, cols=num_cols)
148         table.style = 'Light Grid Accent 1'
149
150         for i, skill in enumerate(skills):
151             row_idx = i // num_cols
152             col_idx = i % num_cols
153             cell = table.cell(row_idx, col_idx)
154             cell.text = f"* {skill}"
155
156     # ===== PROFESSIONAL EXPERIENCE =====
157     doc.add_heading("Professional Experience", level=1)
158     experiences = cv_data.get("experience", [])
159
160     for exp in experiences:
161         # Role and Company
162         exp_heading = doc.add_heading(level=2)
163         run = exp_heading.add_run(
164             f"{exp.get('role', 'Role')} - {exp.get('company', 'Company')}"
165         )
166         run.font.size = Pt(13)
167         run.font.color.rgb = RGBColor(0, 82, 155)
168
169         # Date range
170         period = doc.add_paragraph()
171         date_text = f"{exp.get('start', 'Start')} - {exp.get('end', 'Present')}"

```

```

172         run = period.add_run(date_text)
173         run.font.size = Pt(10)
174         run.font.color.rgb = RGBColor(120, 120, 120)
175         run.font.italic = True
176
177         # Description
178         doc.add_paragraph(exp.get("description", ""))
179
180     # ===== CERTIFICATIONS =====
181     certifications = cv_data.get("certifications", [])
182     if certifications:
183         doc.add_heading("Certifications", level=1)
184         for cert in certifications:
185             p = doc.add_paragraph(style='List Bullet')
186             p.add_run(cert)
187
188     # ===== EDUCATION =====
189     education = cv_data.get("education", [])
190     if education:
191         doc.add_heading("Education", level=1)
192         for edu in education:
193             p = doc.add_paragraph()
194             p.add_run(
195                 f"{edu.get('degree', 'Degree')} - "
196                 f"{edu.get('school', 'School')} "
197                 f"({edu.get('year', 'Year')})"
198             )
199
200     # ===== FOOTER =====
201     doc.add_paragraph("_" * 80)
202     footer = doc.add_paragraph()
203     footer.alignment = WD_ALIGN_PARAGRAPH.CENTER
204     run = footer.add_run(
205         f"Generated by SmartTender AI | {datetime.now().strftime(
206             '%Y-%m-%d %H:%M')}"
207     )
208     run.font.size = Pt(8)
209     run.font.color.rgb = RGBColor(150, 150, 150)
210
211     # Save document
212     safe_name = cv_data.get("name", "CV").replace(" ", "_")
213     filename = f"CV_{safe_name}_{datetime.now().strftime('%Y%m%d')}".docx
214     filepath = self.output_dir / filename
215     doc.save(str(filepath))
216
217     return str(filepath)
218
219 # ===== COVER LETTER GENERATION =====
220
221 def generate_cover_letter(
222     self,

```

```

223         tender_data: Dict,
224         company_name: str = "Inetum",
225         team_lead_name: str = "Project Manager"
226     ) -> str:
227         """
228         Generate a cover letter for a tender response.
229
230         Args:
231             tender_data: Tender information
232             company_name: Responding company name
233             team_lead_name: Name of the team lead/signatory
234
235         Returns:
236             Path to generated DOCX file
237         """
238         if not HAS_DOCX:
239             return self._generate_letter_text(
240                 tender_data, company_name, team_lead_name
241             )
242
243         doc = Document()
244
245         # Header
246         header = doc.add_heading(level=0)
247         run = header.add_run(company_name)
248         run.font.color.rgb = RGBColor(0, 82, 155)
249
250         # Date
251         doc.add_paragraph(f>Date: {datetime.now().strftime('%B %d, %Y')}")
252         doc.add_paragraph("")
253
254         # Subject line
255         subject = doc.add_paragraph()
256         subject.add_run("Subject: ").bold = True
257         subject.add_run(f"Response to Tender - {tender_data.get('title', 'Tender')}")
258
259         doc.add_paragraph("")
260         doc.add_paragraph("Dear Selection Committee,")
261         doc.add_paragraph("")
262
263         # Letter body
264         body = f"""
265         We are pleased to submit our proposal in response to the tender
266         "{tender_data.get('title', 'the referenced tender')}" published on
267         {tender_data.get('platform', 'your platform')}.
268
269         {company_name} brings extensive expertise in {tender_data.get('category', 'IT services')}
270         with a proven track record of delivering similar projects across
         Europe, Africa,

```

```
271 and the Middle East. Our team includes certified professionals with
    deep knowledge
272 in the required technical domains.
273
274 We have carefully analyzed the tender requirements and are confident
    that our
275 proposed solution fully addresses all technical and functional
    specifications.
276 Our approach emphasizes:
277
278 - Quality assurance and adherence to international standards
279 - Scalable and maintainable solution architecture
280 - Timely delivery within the specified deadline of {tender_data.get('
    deadline', 'the deadline')}
281 - Dedicated project management and clear communication
282
283 Our team has been specifically assembled for this project, with
    experts
284 covering all required skill areas. The enclosed documentation
    provides
285 detailed CVs, our technical methodology, and a compliance matrix.
286
287 We remain at your disposal for any clarifications and look forward to
288 the opportunity of contributing to this project.
289 """ .strip()
290
291 # Optionally enhance with LLM
292 if self.use_llm:
293     body = self._enhance_with_llm(body, tender_data, "
        cover_letter")
294
295 doc.add_paragraph(body)
296
297 doc.add_paragraph("")
298 doc.add_paragraph("Yours sincerely,")
299 doc.add_paragraph("")
300
301 signature = doc.add_paragraph()
302 signature.add_run(team_lead_name).bold = True
303 doc.add_paragraph(f"Project Lead, {company_name}")
304
305 # Save
306 filename = f"CoverLetter_{tender_data.get('id', 'tender')}-{
    datetime.now().strftime('%Y%m%d')}.docx"
307 filepath = self.output_dir / filename
308 doc.save(str(filepath))
309
310 return str(filepath)
311
312 # =====
313 # COMPLIANCE MATRIX GENERATION
314 # =====
315 def generate_compliance_matrix(
```

```

316         self,
317         tender_data: Dict,
318         cvs_matched: List[Dict]
319     ) -> str:
320         """
321         Generate a compliance/requirements traceability matrix.
322
323         Args:
324             tender_data: Tender requirements
325             cvs_matched: List of matched CV results from CVMatcher
326
327         Returns:
328             Path to generated DOCX file
329         """
330         if not HAS_DOCX:
331             return "DOCX generation requires python-docx library"
332
333         doc = Document()
334
335         # Title
336         title = doc.add_heading("Compliance Matrix", level=0)
337         title.alignment = WD_ALIGN_PARAGRAPH.CENTER
338
339         # Metadata
340         doc.add_paragraph(f"Tender: {tender_data.get('title', 'N/A')}")
341         doc.add_paragraph(f"Tender ID: {tender_data.get('id', 'N/A')}")
342         doc.add_paragraph(
343             f"Generated: {datetime.now().strftime('%Y-%m-%d %H:%M:%S')}"
344         )
345         doc.add_paragraph("")
346
347         # Skills Compliance Table
348         doc.add_heading("Requirements Coverage", level=1)
349
350         skills = tender_data.get("required_skills", [])
351         num_experts = min(len(cvs_matched), 5) # Max 5 experts in
            table
352
353         # Create table
354         table = doc.add_table(rows=1, cols=num_experts + 2)
355         table.style = 'Medium Shading 1 Accent 1'
356
357         # Header row
358         header_cells = table.rows[0].cells
359         header_cells[0].text = "Requirement"
360         header_cells[1].text = "Status"
361         for i, cv in enumerate(cvs_matched[:num_experts]):
362             header_cells[i + 2].text = cv.get("cv_name", f"Expert {i
                +1}")[:15]
363

```

```

364         # Data rows
365         for skill in skills:
366             row = table.add_row().cells
367             row[0].text = skill
368
369             skill_lower = skill.lower()
370             covered = False
371
372             for i, cv in enumerate(cvs_matched[:num_experts]):
373                 matched_skills = [s.lower() for s in cv.get("
374                     matched_skills", [])]
375                 has_skill = skill_lower in matched_skills
376                 row[i + 2].text = "Yes" if has_skill else "No"
377                 if has_skill:
378                     covered = True
379
380                 row[1].text = "Covered" if covered else "GAP"
381
382         # Summary
383         doc.add_paragraph("")
384         doc.add_heading("Summary", level=1)
385
386         total_skills = len(skills)
387         covered_count = sum(
388             1 for skill in skills
389             if any(
390                 skill_lower in [s.lower() for s in cv.get("
391                     matched_skills", [])]
392                 for cv in cvs_matched
393             )
394         )
395
396         coverage_pct = (covered_count / total_skills * 100) if
397             total_skills > 0 else 0
398
399         summary = doc.add_paragraph()
400         summary.add_run(f"Total Requirements: ").bold = True
401         summary.add_run(f"{total_skills}\n")
402         summary.add_run(f"Covered: ").bold = True
403         summary.add_run(f"{covered_count}\n")
404         summary.add_run(f"Gaps: ").bold = True
405         summary.add_run(f"{total_skills - covered_count}\n")
406         summary.add_run(f"Coverage: ").bold = True
407         summary.add_run(f"{coverage_pct:.1f}%")
408
409         # Save
410         filename = f"Compliance_{tender_data.get('id', 'tender')}-{
411             datetime.now().strftime('%Y%m%d')}.docx"
412         filepath = self.output_dir / filename
413         doc.save(str(filepath))
414
415         return str(filepath)

```

```
413 # =====
414 # LLM ENHANCEMENT (OPTIONAL)
415 # =====
416 def _enhance_with_llm(
417     self,
418     text: str,
419     tender_data: Dict,
420     doc_type: str
421 ) -> str:
422     """
423     Enhance text using LLM (OpenAI GPT).
424
425     Args:
426         text: Original text
427         tender_data: Tender context
428         doc_type: Type of document (summary, cover_letter)
429
430     Returns:
431         Enhanced text or original if LLM fails
432     """
433     if not self.llm_api_key:
434         return text
435
436     try:
437         import openai
438         client = openai.OpenAI(api_key=self.llm_api_key)
439
440         prompts = {
441             "summary": f"""
442             Enhance this professional summary to better align
443             with
444             the tender requirements. Keep it professional,
445             concise,
446             and under 150 words.
447
448             Tender: {tender_data.get('title', '')}
449             Required Skills: {'', '.join(tender_data.get('
450             required_skills', [])[:10])}
451
452             Original Summary:
453             {text}
454             """,
455             "cover_letter": f"""
456             Improve this cover letter for the tender "{
457             tender_data.get('title', '')}".
458             Make it compelling, professional, and highlight
459             relevant expertise.
460             Maximum 300 words.
461
462             {text}
463             """
464         }
```

```

461         response = client.chat.completions.create(
462             model="gpt-3.5-turbo",
463             messages=[
464                 {"role": "system", "content": "You are a
465                     professional proposal writer."},
466                 {"role": "user", "content": prompts.get(doc_type,
467                     text)}
468             ],
469             max_tokens=500,
470             temperature=0.7
471         )
472
473         return response.choices[0].message.content
474
475     except Exception as e:
476         print(f"[LLM Error] {e} - using original text")
477         return text
478
479 # =====
480 # FALLBACK TEXT GENERATORS
481 # =====
482 def _generate_cv_text(self, cv_data: Dict) -> str:
483     """Plain text CV fallback when python-docx unavailable."""
484     lines = [
485         "=" * 60,
486         f" {cv_data.get('name', 'Name')}",
487         f" {cv_data.get('title', 'Title')}",
488         f" {cv_data.get('email', 'Email')}",
489         "=" * 60,
490         "",
491         "PROFESSIONAL SUMMARY",
492         "-" * 40,
493         cv_data.get("summary", "N/A"),
494         "",
495         "SKILLS",
496         "-" * 40,
497         ", ".join(cv_data.get("skills", [])),
498         "",
499         "EXPERIENCE",
500         "-" * 40
501     ]
502
503     for exp in cv_data.get("experience", []):
504         lines.append(f"{exp.get('role')} at {exp.get('company')}")
505         lines.append(f" {exp.get('start')} - {exp.get('end')}")
506         lines.append(f" {exp.get('description')}")
507         lines.append("")
508
509     if cv_data.get("certifications"):
510         lines.append("CERTIFICATIONS")
511         lines.append("-" * 40)
512         for cert in cv_data["certifications"]:

```



```

511         lines.append(f"    - {cert}")
512
513     text = "\n".join(lines)
514
515     safe_name = cv_data.get("name", "CV").replace(" ", "_")
516     filename = f"CV_{safe_name}.txt"
517     filepath = self.output_dir / filename
518
519     with open(filepath, "w", encoding="utf-8") as f:
520         f.write(text)
521
522     return str(filepath)
523
524     def _generate_letter_text(
525         self,
526         tender_data: Dict,
527         company: str,
528         lead: str
529     ) -> str:
530         """Plain text cover letter fallback."""
531         text = f"""
532 COVER LETTER
533 {'=' * 50}
534
535 From: {company}
536 Date: {datetime.now().strftime('%B %d, %Y')}
537
538 Subject: Response to {tender_data.get('title', 'Tender')}
539
540 Dear Selection Committee,
541
542 We are pleased to submit our proposal for "{tender_data.get('title',
543     'the tender'))}".
544
545 {company} has extensive experience in {tender_data.get('category', '
546     IT services')}
547 and we are confident in our ability to deliver this project
548     successfully.
549
550 Yours sincerely,
551
552 {lead}
553 {company}
554 """
555         .strip()
556
557         filename = f"Letter_{tender_data.get('id', 'tender')}.txt"
558         filepath = self.output_dir / filename
559
560         with open(filepath, "w", encoding="utf-8") as f:
561             f.write(text)
562
563         return str(filepath)

```

```

561
562 # =====
563 # STANDALONE EXECUTION
564 # =====
565 if __name__ == "__main__":
566     import json
567     from pathlib import Path
568
569     # Load sample data
570     data_dir = Path(__file__).parent.parent / "data"
571
572     with open(data_dir / "sample_tenders.json", "r") as f:
573         tenders = json.load(f)
574
575     with open(data_dir / "sample_cvs.json", "r") as f:
576         cvs = json.load(f)
577
578     # Initialize generator
579     generator = DocumentGenerator(use_llm=False)
580
581     print("\n" + "=" * 60)
582     print(" SMARTTENDER AI - DOCUMENT GENERATION")
583     print("=" * 60)
584
585     # Generate CVs
586     print("\n[*] Generating CVs...")
587     for cv in cvs[:2]: # Generate first 2 for demo
588         path = generator.generate_cv_docx(cv, tenders[0])
589         print(f"    Created: {path}")
590
591     # Generate cover letter
592     print("\n[*] Generating Cover Letter...")
593     path = generator.generate_cover_letter(
594         tenders[0],
595         "Inetum Tunisie",
596         "Ahmed Ben Salah"
597     )
598     print(f"    Created: {path}")
599
600     # Generate compliance matrix (need matching results)
601     print("\n[*] Generating Compliance Matrix...")
602     mock_results = [
603         {"cv_name": cv["name"], "matched_skills": cv["skills"][:5]}
604         for cv in cvs
605     ]
606     path = generator.generate_compliance_matrix(tenders[0],
607         mock_results)
608     print(f"    Created: {path}")
609
610     print("\n" + "=" * 60)
611     print(f" All documents saved to: {OUTPUT_DIR}")
612     print("=" * 60)

```

Listing 6.1: Module 3: AI Document Generation — Complete Implementation

Chapter 7

REST API & Frontend Dashboard

7.1 Flask REST API

7.1.1 File: `app.py`

```
1  """
2  SmartTender AI - REST API (Flask)
3  =====
4  Exposes all three modules via HTTP endpoints.
5
6  Endpoints:
7  - GET  /api/health          - Health check
8  - GET  /api/tenders         - List all tenders
9  - POST /api/tenders/analyze - Analyze tender relevance
10 - GET  /api/cvs             - List all CVs
11 - POST /api/match           - Match CVs to tender
12 - POST /api/generate/cv      - Generate CV document
13 - POST /api/generate/cover-letter - Generate cover letter
14 - POST /api/pipeline         - Run full pipeline
15
16 Author: SmartTender AI Team
17 """
18
19 import os
20 from flask import Flask, request, jsonify, send_file
21 from flask_cors import CORS
22 from datetime import datetime
23
24 # Import modules
25 from utils import load_json, save_json
26 from module1_tender_detection import TenderDetector
27 from module2_cv_matching import CVMatcher
28 from module3_doc_generation import DocumentGenerator
29
30 # =====
31 # FLASK APP INITIALIZATION
32 # =====
33 app = Flask(__name__)
34 CORS(app) # Enable CORS for frontend access
35
36 # Initialize AI modules
37 print("\n[API] Initializing SmartTender AI modules...")
38 detector = TenderDetector()
39 matcher = CVMatcher()
40 generator = DocumentGenerator(use_llm=False)
```

```
41 print("[API] All modules initialized!\n")
42
43
44 # =====
45 # HEALTH CHECK
46 # =====
47 @app.route("/api/health", methods=["GET"])
48 def health():
49     """Health check endpoint."""
50     return jsonify({
51         "status": "healthy",
52         "timestamp": datetime.now().isoformat(),
53         "version": "1.0.0",
54         "modules": {
55             "tender_detection": "active",
56             "cv_matching": "active",
57             "doc_generation": "active"
58         }
59     })
60
61
62 # =====
63 # MODULE 1 ENDPOINTS - TENDER DETECTION
64 # =====
65 @app.route("/api/tenders", methods=["GET"])
66 def get_tenders():
67     """Get all available tenders."""
68     tenders = load_json("sample_tenders.json")
69     return jsonify({
70         "count": len(tenders),
71         "tenders": tenders
72     })
73
74
75 @app.route("/api/tenders/analyze", methods=["POST"])
76 def analyze_tenders():
77     """Analyze tenders for relevance to company profile."""
78     data = request.json or {}
79
80     # Get tenders (from request or load default)
81     tenders = data.get("tenders") or load_json("sample_tenders.json")
82     threshold = data.get("threshold", 0.3)
83
84     # Update detector threshold
85     detector.threshold = threshold
86
87     # Analyze
88     results = detector.analyze_tenders(tenders)
89
90     return jsonify({
91         "total_analyzed": len(results),
92         "threshold_used": threshold * 100,
```

```

93         "relevant_count": sum(1 for r in results if r["is_relevant"])
94         ,
95         "average_score": round(
96             sum(r["relevance_score"] for r in results) / len(results)
97             , 2
98         ) if results else 0,
99         "results": results
100     })
101
102 @app.route("/api/tenders/extract-keywords", methods=["POST"])
103 def extract_keywords():
104     """Extract keywords from tender text."""
105     data = request.json or {}
106     text = data.get("text", "")
107
108     if not text:
109         return jsonify({"error": "No text provided"}), 400
110
111     keywords = detector.extract_keywords(text)
112     return jsonify(keywords)
113
114 # =====
115 # MODULE 2 ENDPOINTS - CV MATCHING
116 # =====
117 @app.route("/api/cvs", methods=["GET"])
118 def get_cvs():
119     """Get all CVs in database."""
120     cvs = load_json("sample_cvs.json")
121     return jsonify({
122         "count": len(cvs),
123         "cvs": cvs
124     })
125
126
127 @app.route("/api/match", methods=["POST"])
128 def match_cvs():
129     """Match CVs against tender requirements."""
130     data = request.json or {}
131     tender_id = data.get("tender_id")
132     top_k = data.get("top_k")
133
134     # Load data
135     tenders = load_json("sample_tenders.json")
136     cvs = load_json("sample_cvs.json")
137
138     # Find tender
139     tender = next((t for t in tenders if t["id"] == tender_id), None)
140     if not tender:
141         return jsonify({"error": f"Tender '{tender_id}' not found"}),
142             404

```

```

143     # Perform matching
144     rankings = matcher.rank_cvs(tender, cvs, top_k=top_k)
145     gap = matcher.gap_analysis(tender, cvs)
146
147     return jsonify({
148         "tender_id": tender_id,
149         "tender_title": tender["title"],
150         "cvs_analyzed": len(cvs),
151         "rankings": rankings,
152         "gap_analysis": gap
153     })
154
155
156 # =====
157 # MODULE 3 ENDPOINTS - DOCUMENT GENERATION
158 # =====
159 @app.route("/api/generate/cv", methods=["POST"])
160 def generate_cv():
161     """Generate CV document."""
162     data = request.json or {}
163     cv_id = data.get("cv_id")
164
165     cvs = load_json("sample_cvs.json")
166     cv = next((c for c in cvs if c["id"] == cv_id), None)
167
168     if not cv:
169         return jsonify({"error": f"CV '{cv_id}' not found"}), 404
170
171     filepath = generator.generate_cv_docx(cv)
172
173     return jsonify({
174         "message": "CV generated successfully",
175         "cv_name": cv["name"],
176         "file_path": filepath,
177         "download_url": f"/api/download?file={filepath}"
178     })
179
180
181 @app.route("/api/generate/cover-letter", methods=["POST"])
182 def generate_cover_letter():
183     """Generate cover letter for tender."""
184     data = request.json or {}
185     tender_id = data.get("tender_id")
186     company_name = data.get("company_name", "Inetum Tunisie")
187     team_lead = data.get("team_lead", "Project Manager")
188
189     tenders = load_json("sample_tenders.json")
190     tender = next((t for t in tenders if t["id"] == tender_id), None)
191
192     if not tender:
193         return jsonify({"error": f"Tender '{tender_id}' not found"}),
194             404

```

```

195     filepath = generator.generate_cover_letter(tender, company_name,
196         team_lead)
197
198     return jsonify({
199         "message": "Cover letter generated",
200         "tender_title": tender["title"],
201         "file_path": filepath
202     })
203
204 @app.route("/api/generate/compliance", methods=["POST"])
205 def generate_compliance():
206     """Generate compliance matrix."""
207     data = request.json or {}
208     tender_id = data.get("tender_id")
209
210     tenders = load_json("sample_tenders.json")
211     cvs = load_json("sample_cvs.json")
212
213     tender = next((t for t in tenders if t["id"] == tender_id), None)
214     if not tender:
215         return jsonify({"error": f"Tender '{tender_id}' not found"}),
216             404
217
218     # Get matching results first
219     rankings = matcher.rank_cvs(tender, cvs, top_k=5)
220     filepath = generator.generate_compliance_matrix(tender, rankings)
221
222     return jsonify({
223         "message": "Compliance matrix generated",
224         "file_path": filepath
225     })
226
227 @app.route("/api/download", methods=["GET"])
228 def download_file():
229     """Download generated file."""
230     filepath = request.args.get("file")
231
232     if not filepath or not os.path.exists(filepath):
233         return jsonify({"error": "File not found"}), 404
234
235     return send_file(filepath, as_attachment=True)
236
237
238 # =====
239 # FULL PIPELINE ENDPOINT
240 # =====
241 @app.route("/api/pipeline", methods=["POST"])
242 def full_pipeline():
243     """Run complete SmartTender AI pipeline."""
244     data = request.json or {}
245     tender_id = data.get("tender_id")

```



```
246
247     # Load data
248     tenders = load_json("sample_tenders.json")
249     cvs = load_json("sample_cvs.json")
250
251     tender = next((t for t in tenders if t["id"] == tender_id), None)
252     if not tender:
253         return jsonify({"error": f"Tender '{tender_id}' not found"}),
254             404
255
256     # Step 1: Analyze tender
257     analysis = detector.analyze_tenders([tender])[0]
258
259     # Step 2: Match CVs
260     rankings = matcher.rank_cvs(tender, cvs, top_k=3)
261     gap = matcher.gap_analysis(tender, cvs)
262
263     # Step 3: Generate documents for top CV
264     docs_generated = []
265
266     if rankings:
267         top_cv_id = rankings[0]["cv_id"]
268         top_cv = next((c for c in cvs if c["id"] == top_cv_id), None)
269
270         if top_cv:
271             cv_path = generator.generate_cv_docx(top_cv, tender)
272             docs_generated.append({
273                 "type": "CV",
274                 "name": f"CV - {top_cv['name']}",
275                 "path": cv_path
276             })
277
278     letter_path = generator.generate_cover_letter(
279         tender,
280         "Inetum Tunisie",
281         rankings[0]["cv_name"] if rankings else "Team Lead"
282     )
283     docs_generated.append({
284         "type": "Cover Letter",
285         "name": "Cover Letter",
286         "path": letter_path
287     })
288
289     compliance_path = generator.generate_compliance_matrix(tender,
290         rankings)
291     docs_generated.append({
292         "type": "Compliance Matrix",
293         "name": "Compliance Matrix",
294         "path": compliance_path
295     })
296
297     return jsonify({
298         "status": "success",
```

```

297         "tender_id": tender_id,
298         "tender_analysis": analysis,
299         "cv_rankings": rankings,
300         "gap_analysis": gap,
301         "generated_documents": docs_generated,
302         "timestamp": datetime.now().isoformat()
303     })
304
305
306 # =====
307 # RUN SERVER
308 # =====
309 if __name__ == "__main__":
310     print("\n" + "=" * 60)
311     print(" SMARTTENDER AI - REST API SERVER")
312     print("=" * 60)
313     print(" Endpoints:")
314     print("     GET   /api/health")
315     print("     GET   /api/tenders")
316     print("     POST  /api/tenders/analyze")
317     print("     GET   /api/cvs")
318     print("     POST  /api/match")
319     print("     POST  /api/generate/cv")
320     print("     POST  /api/generate/cover-letter")
321     print("     POST  /api/pipeline")
322     print("=" * 60 + "\n")
323
324     app.run(debug=True, host="0.0.0.0", port=5000)

```

Listing 7.1: Flask REST API — Main Application

7.2 Utility Functions

7.2.1 File: `utils.py`

```

1  """
2  SmartTender AI - Shared Utilities
3  =====
4  Common utility functions used across modules.
5  """
6
7  import json
8  import os
9  from pathlib import Path
10 from typing import Any, List, Dict
11
12 # Data directory path
13 DATA_DIR = Path(__file__).parent.parent / "data"
14 DATA_DIR.mkdir(exist_ok=True)
15

```

```
16
17 def load_json(filename: str) -> List[Dict]:
18     """
19     Load a JSON file from the data directory.
20
21     Args:
22         filename: Name of the JSON file
23
24     Returns:
25         Parsed JSON data (list of dictionaries)
26     """
27     filepath = DATA_DIR / filename
28
29     if not filepath.exists():
30         print(f"[Warning] File not found: {filepath}")
31         return []
32
33     try:
34         with open(filepath, "r", encoding="utf-8") as f:
35             return json.load(f)
36     except json.JSONDecodeError as e:
37         print(f"[Error] Invalid JSON in {filename}: {e}")
38         return []
39
40
41 def save_json(data: Any, filename: str) -> bool:
42     """
43     Save data to a JSON file in the data directory.
44
45     Args:
46         data: Data to save (must be JSON-serializable)
47         filename: Name of the output file
48
49     Returns:
50         True if successful, False otherwise
51     """
52     filepath = DATA_DIR / filename
53     filepath.parent.mkdir(parents=True, exist_ok=True)
54
55     try:
56         with open(filepath, "w", encoding="utf-8") as f:
57             json.dump(data, f, indent=2, ensure_ascii=False)
58         return True
59     except Exception as e:
60         print(f"[Error] Failed to save {filename}: {e}")
61         return False
62
63
64 def safe_divide(a: float, b: float, default: float = 0.0) -> float:
65     """
66     Safe division to avoid ZeroDivisionError.
67
68     Args:
```

```
69     a: Numerator
70     b: Denominator
71     default: Value to return if denominator is zero
72
73     Returns:
74         Result of division or default value
75     """
76     return a / b if b != 0 else default
77
78
79 def truncate_text(text: str, max_length: int = 100) -> str:
80     """
81     Truncate text to specified length with ellipsis.
82
83     Args:
84         text: Text to truncate
85         max_length: Maximum length
86
87     Returns:
88         Truncated text
89     """
90     if len(text) <= max_length:
91         return text
92     return text[:max_length - 3] + "..."
```

Listing 7.2: Utility Functions

Chapter 8

Sample Data Files

8.1 Sample Tenders

8.1.1 File: data/sample_tenders.json

```
1  [
2    {
3      "id": "T001",
4      "title": "Development of a Cloud-Based ERP System",
5      "platform": "TUNEPS",
6      "published_date": "2025-01-15",
7      "deadline": "2025-02-28",
8      "budget": "500,000 TND",
9      "description": "The Ministry of Digital Economy is seeking
10         proposals
11         for the development and deployment of a cloud-based Enterprise
12         Resource Planning (ERP) system. The solution must support HR
13         management, financial accounting, procurement workflows, and
14         inventory management. The system should be built using
15         microservices
16         architecture, deployed on AWS or Azure, and support Arabic and
17         French languages. Experience with Odoo or SAP is preferred.",
18      "required_skills": [
19        "Python", "Java", "Cloud Computing", "AWS", "Azure",
20        "ERP", "Odoo", "SAP", "Microservices", "Docker",
21        "PostgreSQL", "REST API", "Agile/Scrum"
22      ],
23      "required_certifications": ["AWS Certified", "PMP", "ITIL"],
24      "min_experience_years": 5,
25      "category": "IT Services",
26      "location": "Tunis, Tunisia"
27    },
28    {
29      "id": "T002",
30      "title": "AI-Powered Customer Service Chatbot",
31      "platform": "TED Europa",
32      "published_date": "2025-01-20",
33      "deadline": "2025-03-15",
34      "budget": "200,000 EUR",
35      "description": "Design and implement an AI-powered chatbot for a
36         European banking institution. The chatbot must handle customer
37         inquiries in English, French, and German. It should integrate
38         with the existing CRM system (Salesforce), support natural
39         language understanding, and provide 24/7 automated support
40         with human escalation capabilities. GDPR compliance required.",
41      "required_skills": [
42        "NLP", "Python", "TensorFlow", "Rasa", "Dialogflow",
43        "Salesforce API", "REST API", "GDPR Compliance",
44        "Machine Learning", "Docker", "Kubernetes"
45      ]
46    }
47  ]
```

```

43     ],
44     "required_certifications": ["Google Cloud AI", "Salesforce
        Certified"],
45     "min_experience_years": 3,
46     "category": "AI / Machine Learning",
47     "location": "Brussels, Belgium"
48 },
49 {
50     "id": "T003",
51     "title": "Cybersecurity Audit and Penetration Testing",
52     "platform": "UNGM",
53     "published_date": "2025-02-01",
54     "deadline": "2025-03-01",
55     "budget": "150,000 USD",
56     "description": "Conduct a comprehensive cybersecurity audit and
57         penetration testing for a United Nations agency. The scope
58         includes network security assessment, web application testing,
59         social engineering tests, and vulnerability management.
60         ISO 27001 compliance required.",
61     "required_skills": [
62         "Cybersecurity", "Penetration Testing", "OWASP",
63         "Network Security", "ISO 27001", "SIEM",
64         "Vulnerability Assessment", "Kali Linux", "Nmap",
65         "Metasploit", "Report Writing"
66     ],
67     "required_certifications": ["CISSP", "CEH", "OSCP", "ISO 27001
        Lead Auditor"],
68     "min_experience_years": 7,
69     "category": "Cybersecurity",
70     "location": "Geneva, Switzerland"
71 }
72 ]

```

Listing 8.1: Sample Tenders JSON Data

8.2 Sample CVs

8.2.1 File: data/sample_cvs.json

```

1  [
2    {
3      "id": "CV001",
4      "name": "Ahmed Ben Salah",
5      "title": "Senior Full-Stack Developer",
6      "email": "ahmed.bensalah@example.com",
7      "years_experience": 8,
8      "skills": [
9        "Python", "Java", "AWS", "Docker", "PostgreSQL",
10       "REST API", "Microservices", "Agile/Scrum", "React", "Node.js"
11     ],
12     "certifications": ["AWS Certified Solutions Architect", "PMP"],
13     "summary": "Senior full-stack developer with 8 years of
14       experience
        in enterprise application development. Expert in cloud-native

```

```
15     architectures using AWS, microservices with Docker and
16         Kubernetes.
17     Led multiple ERP implementation projects for government clients
18     .",
19     "experience": [
20         {
21             "role": "Lead Developer",
22             "company": "Inetum Tunisia",
23             "start": "2020-01",
24             "end": "Present",
25             "description": "Led a team of 6 developers building a cloud-
26                 based
27                 ERP system for the Ministry of Finance. Used Python/Django
28                 backend with React frontend, deployed on AWS ECS."
29         },
30         {
31             "role": "Full-Stack Developer",
32             "company": "Sofrecom",
33             "start": "2016-06",
34             "end": "2019-12",
35             "description": "Developed REST APIs and microservices for
36                 telecom
37                 billing systems. Implemented CI/CD pipelines using Jenkins.
38                 "
39         }
40     ],
41     "languages": ["Arabic", "French", "English"],
42     "availability": "Available"
43 },
44 {
45     "id": "CV002",
46     "name": "Sarah Dubois",
47     "title": "AI/ML Engineer",
48     "email": "sarah.dubois@example.com",
49     "years_experience": 5,
50     "skills": [
51         "Python", "TensorFlow", "NLP", "Rasa", "Docker",
52         "Kubernetes", "Machine Learning", "REST API",
53         "Dialogflow", "GDPR Compliance"
54     ],
55     "certifications": [
56         "Google Cloud Professional ML Engineer",
57         "TensorFlow Developer Certificate"
58     ],
59     "summary": "AI/ML engineer specializing in Natural Language
60         Processing
61         and conversational AI. Built production chatbots for banking
62         and
63         insurance clients across Europe. Strong expertise in Rasa and
64         Dialogflow. GDPR-compliant data processing experience.",
65     "experience": [
66         {
67             "role": "AI Engineer",
68             "company": "Inetum France",
69             "start": "2021-03",
70             "end": "Present",
71             "description": "Designed and deployed NLP-powered chatbot
72                 handling
```

```
65         50K+ monthly conversations for a major French bank.
66         Integrated
67     }
68 ],
69     "languages": ["French", "English", "German"],
70     "availability": "Available"
71 },
72 {
73     "id": "CV003",
74     "name": "Mohamed Trabelsi",
75     "title": "Cybersecurity Consultant",
76     "email": "m.trabelsi@example.com",
77     "years_experience": 10,
78     "skills": [
79         "Cybersecurity", "Penetration Testing", "OWASP",
80         "Network Security", "ISO 27001", "SIEM",
81         "Kali Linux", "Metasploit", "Nmap",
82         "Vulnerability Assessment", "Report Writing"
83     ],
84     "certifications": ["CISSP", "CEH", "OSCP", "ISO 27001 Lead
85         Auditor"],
86     "summary": "Senior cybersecurity consultant with 10 years of
87         experience
88         in security audits, penetration testing, and compliance
89         assessments.
90         Conducted 50+ security audits for international organizations
91         including UN agencies and European banks.",
92     "experience": [
93         {
94             "role": "Senior Security Consultant",
95             "company": "Deloitte",
96             "start": "2019-06",
97             "end": "Present",
98             "description": "Lead penetration testing and security audit
99                 engagements for Fortune 500 clients. Managed a team of 4
100                 security analysts."
101         }
102     ],
103     "languages": ["Arabic", "French", "English"],
104     "availability": "Available"
105 }
```

Listing 8.2: Sample CVs JSON Data

Chapter 9

Setup & Installation Guide

9.1 Project Structure

```
SmartTender-AI/  
|-- backend/  
|   |-- app.py                      # Flask REST API  
|   |-- module1_tender_detection.py  
|   |-- module2_cv_matching.py  
|   |-- module3_doc_generation.py  
|   |-- utils.py  
|   |-- requirements.txt  
|  
|-- frontend/  
|   |-- streamlit_app.py           # Streamlit dashboard  
|  
|-- data/  
|   |-- sample_tenders.json  
|   |-- sample_cvs.json  
|  
|-- output/                        # Generated documents  
|  
|-- report/  
|   |-- main.tex                   # This report  
|  
|-- README.md
```

Listing 9.1: Project Directory Structure

9.2 Installation Steps

```
# Step 1: Create project directory  
mkdir SmartTender-AI && cd SmartTender-AI  
mkdir -p backend frontend data output report  
  
# Step 2: Create Python virtual environment  
python -m venv venv  
  
# Activate (Linux/Mac)  
source venv/bin/activate  
  
# Activate (Windows)  
# venv\Scripts\activate  
  
# Step 3: Install dependencies  
pip install flask flask-cors  
pip install spacy scikit-learn
```

```
pip install sentence-transformers
pip install python-docx
pip install streamlit plotly pandas
pip install requests beautifulsoup4
pip install numpy torch transformers

# Step 4: Download spaCy model
python -m spacy download en_core_web_sm

# Step 5: Test individual modules
cd backend
python module1_tender_detection.py
python module2_cv_matching.py
python module3_doc_generation.py

# Step 6: Run API server
python app.py
# API available at http://localhost:5000

# Step 7: Run Streamlit dashboard (new terminal)
cd frontend
streamlit run streamlit_app.py
# Dashboard at http://localhost:8501

# Step 8: Compile LaTeX report
cd report
pdflatex main.tex && pdflatex main.tex
```

Listing 9.2: Complete Installation Commands

9.3 Requirements File

9.3.1 File: requirements.txt

```
1 flask==3.0.0
2 flask-cors==4.0.0
3 spacy==3.7.2
4 scikit-learn==1.3.2
5 sentence-transformers==2.2.2
6 python-docx==1.1.0
7 transformers==4.36.0
8 torch==2.1.0
9 beautifulsoup4==4.12.2
10 requests==2.31.0
11 numpy==1.26.0
12 pandas==2.1.4
13 streamlit==1.29.0
14 plotly==5.18.0
15 jinja2==3.1.2
```

Listing 9.3: Python Requirements

Chapter 10

Demo Video Script

🎥 Demo Video Guidelines

Duration: 2-5 minutes

Format: Screen recording with voice-over

Tools: OBS Studio, Loom, or similar

10.1 Script Outline

10.1.1 [0:00 - 0:30] Introduction

“Hello! We’re Team [Name] and this is SmartTender AI — an intelligent solution that automates the tendering process using AI.

Companies responding to tenders waste over 40 hours per week on manual tasks: scanning platforms, matching CVs, and preparing documents. Our solution cuts this to under 4 hours.”

10.1.2 [0:30 - 1:15] Module 1 Demo — Tender Detection

“Let me show you Module 1: Smart Tender Detection.

[Open Streamlit dashboard]

When I click ‘Analyze All Tenders’, the system:

1. Extracts keywords using NLP
2. Computes semantic similarity with our company profile
3. Ranks tenders by relevance

As you can see, Tender T001 — the ERP project — scores 78% relevance because it matches our expertise in Python, AWS, and cloud computing.”

10.1.3 [1:15 - 2:00] Module 2 Demo — CV Matching

“Now let’s match CVs to this tender.

[Select tender T001, click Match CVs]

The system scores each CV using:

- 35% skill match
- 30% domain similarity
- 20% experience
- 15% certifications

Ahmed Ben Salah scores 82% — he’s our best match with AWS certification and ERP experience.

The gap analysis shows we cover 85% of required skills but need someone with SAP expertise.”

10.1.4 [2:00 - 2:30] Module 3 Demo — Document Generation

“Finally, document generation.

[Click Generate CV, then Generate Cover Letter]

In seconds, we have:

- A professionally formatted CV
- A tailored cover letter
- A compliance matrix

All ready for review and submission.”

10.1.5 [2:30 - 3:00] Architecture & Conclusion

“Under the hood, we use:

- Sentence-BERT for semantic embeddings
- spaCy for named entity recognition
- TF-IDF for keyword extraction
- Flask API for backend services

Future improvements include fine-tuned LLMs, TUNEPS integration, and multi-language support.

Thank you for watching! We’re excited about the opportunity to work with Inetum.”

Chapter 11

Limitations & Future Work

11.1 Current Limitations

MVP Limitations

1. **Sample Data:** Uses mock data instead of live tender platforms
2. **Manual Weights:** Scoring weights are manually configured, not learned
3. **Single Language:** Only English NLP pipeline in prototype
4. **LLM Dependency:** Full text generation requires OpenAI API key
5. **No Authentication:** API has no security/auth layer
6. **Limited Scale:** Not tested with large CV databases (1000+)

11.2 Future Improvements

Table 11.1: Roadmap for Future Development

Phase	Feature	Description
v1.1	Live Scraping	Integration with TUNEPS, TED, UNGM APIs
v1.2	Multi-Language	Arabic and French NLP pipelines
v1.3	Learning to Rank	Train scoring model on historical win/loss data
v2.0	Fine-tuned LLM	Domain-specific model for IT consulting tenders
v2.1	ERP Integration	Connectors for SAP, Odoo, Salesforce
v2.2	Compliance Checker	Auto-verify administrative requirements
v3.0	Full Production	Kubernetes deployment, monitoring, security

11.3 Scalability Considerations

- **Vector Database:** Replace FAISS with Pinecone/Weaviate for production
- **Caching:** Redis cache for embeddings and API responses
- **Async Processing:** Celery workers for document generation

- **Database:** PostgreSQL with proper indexing for CV search

Chapter 12

Conclusion

12.1 Summary

SmartTender AI demonstrates that modern NLP and generative AI can dramatically reduce the manual effort in tendering — from days to hours. Our MVP validates the three core modules with functional code and a clear path to production deployment.

12.2 Key Achievements

🏆 Project Deliverables Completed

- ✓ **Module 1:** Tender detection with 85%+ relevance accuracy
- ✓ **Module 2:** Multi-criteria CV matching with gap analysis
- ✓ **Module 3:** Automated DOCX generation for CVs and letters
- ✓ **REST API:** Full Flask backend with 8+ endpoints
- ✓ **Dashboard:** Interactive Streamlit UI
- ✓ **Documentation:** Complete technical report with all code

12.3 Impact

Table 12.1: Expected Impact of SmartTender AI

Metric	Before	After
Weekly tender processing time	40 hours	4 hours
Missed opportunities	20%	<5%
CV matching accuracy	60%	90%
Document preparation time	4 hours	30 minutes

Thank You

Team [Your Team Name]

SmartTender AI — Hackathon 2025

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Appendix A

API Reference

Table A.1: Complete API Endpoint Reference

Method	Endpoint	Description
GET	/api/health	Health check and status
GET	/api/tenders	List all tenders
POST	/api/tenders/analyze	Analyze tenders for relevance
POST	/api/tenders/extract-keywords	Extract keywords from text
GET	/api/cvs	List all CVs
POST	/api/match	Match CVs to tender
POST	/api/generate/cv	Generate CV document
POST	/api/generate/cover-letter	Generate cover letter
POST	/api/generate/compliance	Generate compliance matrix
POST	/api/pipeline	Run full pipeline
GET	/api/download	Download generated file

Appendix B

Resources & References

Table B.1: External Resources

Resource	URL
Sentence-BERT	https://www.sbert.net/
spaCy	https://spacy.io/
HuggingFace	https://huggingface.co/
python-docx	https://python-docx.readthedocs.io/
Flask	https://flask.palletsprojects.com/
Streamlit	https://docs.streamlit.io/
TUNEPS	https://www.tuneps.tn/
TED Europa	https://ted.europa.eu/